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**Movements and Behavior of Smallmouth
Bass, *Micropterus dolomieu*, and Rock Bass,
Ambloplites rupestris, in Southcentral Lake
Ontario and Two Tributaries**

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ABSTRACT

Movements and behavior of 56 stream-spawning smallmouth bass and rock bass from Lake Ontario were examined in spring and summer 1985 and 1986. Fish were captured by electrofishing near spawning areas in two tributaries 60 km apart, fitted with temperature-sensing radiotransmitters and released in their "home" tributaries or displaced to the lake 3-10 km from these tributaries. Twenty-six percent of displaced smallmouth bass and 42% of rock bass returned to their original sites of capture in tributaries. Non-homing fish generally dispersed along the lake shore and eventually established small residence areas in the lake, often near home or other tributaries. No differences were observed in movement rates, temperatures occupied or diel activity patterns between years or tributary stocks, but differences were found between species and between lake and tributary habitats for smallmouth bass. Smallmouth bass homing to, or released in, tributaries generally established home ranges downstream from spawning areas and often occupied tributaries all summer, but rock bass did not. Both species occupied temperatures between 20-22 C in lake and tributary habitats when available. Smallmouth bass were most active in May, least active in July and generally more active in the lake than in tributaries. Rock bass activity did not appear to vary by season or habitat. Both species were most active in Lake Ontario at mid-day; differences in tributaries were less pronounced.

INTRODUCTION

Studies of the movements and distribution of smallmouth bass (*Micropterus dolomieu*) and rock bass (*Ambloplites rupestris*) in the Great Lakes are relatively few and have concentrated primarily on seemingly large lacustrine stocks associated with particular bays or islands (Stone et al. 1954, Fraser 1955, Fry and Watt 1957, Latta 1963, White 1970, Kelso 1973, MacLean and Teleki 1977). In southcentral Lake Ontario these species are scarce by comparison, possibly because the substrate is mostly sand and mud, the shoreline is devoid of islands and there are few bays, shoals or other protective

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shallows (Figure 1). Smallmouth bass and rock bass from the lake congregate in tributaries from April to July to spawn, possibly because streams afford warmer temperatures and protection from wave action. However, the importance of tributaries, as spawning habitat for populations of these centrarchids occupying large lakes or rivers has rarely been addressed (Cuerrier 1943, Webster 1954, Montgomery and Fickeisen 1978). Only one study has described the migratory movements of a stream-spawning lacustrine smallmouth bass stock (Robbins and MacCrimmon 1977) and our study provides the first published report of similar behavior by a lacustrine rock bass stock that we know of.

Examination of the offshore thermal discharge at Somerset Station (Niagara County, NY; Figure 1) and its possible effects on sport fish in Lake Ontario (NYSEG 1987) provided an opportunity to investigate the behavioral ecology of smallmouth bass and rock bass. Here we describe homing of displaced fish as well as movements, activity patterns and temperatures occupied by smallmouth bass and rock bass in southcentral Lake Ontario and two spawning tributaries.

METHODS

From April to September (1985 & 1986), 44 smallmouth bass (31.5-45.0 cm total length; 0.5-1.4 kg) and 12 rock bass (23.5-27.5 cm total length; 0.2-0.4 kg) were captured by electrofishing and fitted with externally mounted (Haynes and Nettles 1983, Mellas and Haynes 1985) temperature-sensing radiotransmitters (sensitivity 0.2-0.5 °C; Ross et al. 1979). In 1985, 28 smallmouth bass were captured in two tributaries: 14 in the first kilometer upstream from Lake Ontario in Sandy Creek (Monroe County, NY; Figure 1) and 14 below Burt Dam 3.5 km upstream from Lake Ontario in Eighteen Mile Creek (Niagara County, NY; Figure 1). Because of the season, size and reproductive condition of fish, it was assumed that smallmouth bass and rock bass were using Sandy and Eighteen Mile Creeks as spawning or "home" waters. Both species were commonly observed nesting in these tributaries during late spring and early summer. Captured fish were released at one of three locations: at capture sites in home tributaries; downstream from capture sites (usually near the mouth of home streams); or in Lake Ontario 3-10 km east or west of home tributaries. In 1986, 16 smallmouth bass and 12 rock bass were captured below Burt Dam in Eighteen Mile Creek. Smallmouth bass were released near the mouth of Eighteen Mile Creek or 10 km east of the creek in Lake Ontario at Somerset Station. All rock bass were released at Somerset Station.

To determine movements in the lake and into streams, fish were tracked by airplane 1-4 times weekly (Winter 1983, Gerber 1987). Maximum flying range was between the Niagara and Genesee Rivers,

less than 2 km offshore. Due to attenuation of radio signals, fish could not be detected more than 12 m deep in Lake Ontario (Haynes, unpublished data).

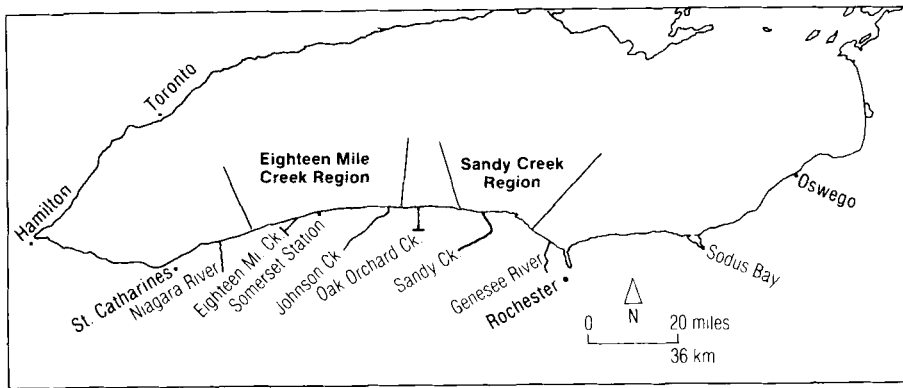


Figure 1. Map of Lake Ontario showing ranges of fish movement in the Sandy and Eighteen Mile Creek regions.

To determine diel activity patterns, temperatures and home ranges, fish in the Sandy and Eighteen Mile Creek regions (Figure 1) were tracked by truck at approximately hourly intervals in separate 5 or 10 h periods for a complete diel cycle once each week from May through August. Fish were observed each week during midnight (2200-0300), dawn (0300-0800), morning (0800-1230), afternoon (1230-1700), and dusk (1700-2200) periods. In 1986, the midnight tracking period was excluded because little activity had been observed during this period in 1985. A fish was considered inactive when there was no variation in radio signal strength, as determined by listening, and no position change greater than 10^0 from the previous observation, as determined by bearings from fixed locations (Winter 1983). Conversely, a fish was considered active when there was a definite variation in signal strength and/or change in position. Depending on the type of analysis, fish activity was expressed as the percentage of active observations per hour, diel period or month.

Estimates of movement rates were made separately for lake and creek habitats in two ways. Seasonal movement rates (km/d) were estimated as the total km traveled divided by the number of days that fish in each habitat remained in radio contact. Short-term movement rates (km/h) were estimated as the distance traveled between consecutive observations less than two hours apart divided by the number of hours between observations. Short-term movement rates were calculated only when displacement was observed and, therefore,

reflect maximum, not average, hourly rates. Distances between consecutive observations were estimated as the shortest water route between locations.

Analyses of seasonal and short-term movement rates (log transformed), temperatures and activity were by oneway and multiway ANOVA (Stats Plus 1982; $\alpha = 0.05$). Comparisons were made by stock (Sandy Creek, Eighteen Mile Creek; smallmouth bass only); by year (1985, 1986; smallmouth bass only); by species (smallmouth bass, rock bass); and by habitat (Lake Ontario, tributaries). Temperature and activity were compared by month and activity was compared by diel period (midnight, dawn, morning, afternoon, dusk). Temperatures were not compared by diel period because complete diel cycle data were available for only a few dates and fish. Also, ambient temperature fluctuations typically were large and obscured any behavioral temperature selection which may have taken place in a given diel period. Simple linear regression was used to compare movement rates (km/d, km/h) with days tracked and fish length. Second-order polynomial regression was used to compare activity with time of day.

Movements of smallmouth bass released at or returning to home tributaries were analyzed for patterns of restricted habitat use or home range. Only fish that occupied tributaries for ten or more days were considered to have established a home range. A center of activity was calculated for each of these fish by averaging the distances of all locations from the tributary mouth. A cumulative frequency distribution of activity radii, or the distance of each location of a fish from the activity center (Dice and Clark 1953), was compiled as described by Siniff and Tester (1965) and Ables (1969). Movements of fish in Lake Ontario were not similarly analyzed because small movements of fish occupying residence areas were often beyond the resolution of equipment used to triangulate positions from shore.

RESULTS

Analysis of seasonal (km/d) and short-term (km/h) movement rates, temperatures, and activity levels of smallmouth bass revealed no significant differences between Sandy and Eighteen Mile Creek stocks in 1985 or between 1985 and 1986 for the Eighteen Mile Creek stock (Table 1). However, differences between lake and tributary habitats were considerable for smallmouth bass, but less so for rock bass (Table 1). Therefore, in subsequent analyses, smallmouth bass data for both years and both stocks were combined, but habitats and species were considered separately (Table 2). Table 3 indicates the last known locations of radiotagged bass.

Table 1. Seasonal movement rates (km/d), short-term movement rates (km/h), seasonal temperatures (C) and seasonal activity levels (% active observations) of smallmouth bass compared by habitat (Lake Ontario vs. tributaries), year (1985 vs. 1986) and stock (Sandy Creek vs. Eighteen Mile Creek) and rock bass compared by habitat (Lake Ontario vs. tributaries).

Category	Seasonal Rate n $\bar{x} \pm$ SD		Short-term Rate n $\bar{x} \pm$ SD		Temperature n $\bar{x} \pm$ SD		Activity n $\bar{x} \pm$ SD	
Smallmouth by Habitat								
Lake Ontario	35	0.7 $\pm$ 0.7	20	0.3 $\pm$ 0.2	35	15.1 $\pm$ 4.0	25	62 $\pm$ 14
Tributaries	20	0.2 $\pm$ 0.2	19	0.1 $\pm$ 0.1	22	18.5 $\pm$ 2.7	18	35 $\pm$ 18
P value	0.002		<0.001		<0.001		<0.001	
Smallmouth by Year								
In Lake Ontario								
1985	21	0.7 $\pm$ 0.7	9	0.3 $\pm$ 0.2	21	15.6 $\pm$ 4.2	15	61 $\pm$ 15
1986	14	0.7 $\pm$ 0.7	11	0.3 $\pm$ 0.2	14	14.3 $\pm$ 3.7	10	64 $\pm$ 14
P Value	>0.5		>0.5		>0.5		>0.5	
In Tributaries								
1985	11	0.2 $\pm$ 0.1	10	0.1 $\pm$ 0.1	13	18.8 $\pm$ 3.1	10	32 $\pm$ 18
1986	9	0.3 $\pm$ 0.2	9	0.1 $\pm$ 0.1	9	17.9 $\pm$ 2.1	8	39 $\pm$ 17
P Value	0.246		>0.5		>0.5		>0.5	
Smallmouth by Stock								
In Lake Ontario								
Eighteen Mi. Cr.	10	0.7 $\pm$ 0.8	1	0.1 $\pm$ 0	10	14.4 $\pm$ 3.5	5	56 $\pm$ 12
Sandy Cr.	11	0.7 $\pm$ 0.6	8	0.3 $\pm$ 0.2	11	16.6 $\pm$ 4.6	10	63 $\pm$ 16
P Value	>0.5		---		0.238		>0.5	
In Tributaries								
Eighteen Mi. Cr.	4	0.2 $\pm$ 0.1	4	0.2 $\pm$ 0.1	5	19.1 $\pm$ 1.9	4	35 $\pm$ 13
Sandy Cr.	7	0.2 $\pm$ 0.1	6	0.1 $\pm$ 0.1	8	18.7 $\pm$ 3.7	6	30 $\pm$ 22
P Value	>0.5		0.270		>0.5		>0.5	
Rock Bass by Habitat								
Lake Ontario	12	1.4 $\pm$ 1.5	11	0.4 $\pm$ 0.2	12	13.7 $\pm$ 4.0	8	62 $\pm$ 19
Tributaries	5	0.5 $\pm$ 0.3	4	0.2 $\pm$ 0.1	5	16.0 $\pm$ 3.2	5	62 $\pm$ 24
P Value	0.223		0.038		0.254		>0.5	

Lake Ontario. Seasonal movement rates by smallmouth bass and rock bass in Lake Ontario averaged 0.7 km/d and 1.4 km/d, respectively (Table 2), but differences between species were largely differences between homing and non-homing fish. Seasonal movement rates were greater ($P < 0.001$) for fish homing to tributaries (smallmouth bass = 1.2 km/d; rock bass = 2.9 km/d) than for non-homing fish that remained in the lake (smallmouth bass = rock bass = 0.6 km/d). Rates were also greater ($P = 0.003$) for homing rock bass (2.9 km/d) than for smallmouth bass (1.2 km/d). Homing smallmouth bass (26%) and rock bass (42%) traveled up to 9.0 km and 11.0 km in a single day, respectively, usually in a direct path back to home tributaries. Short-term movement rates averaged 0.3 km/h in Lake Ontario and did not differ by species (Table 2). Maximum short-term movement rates

for smallmouth bass and rock bass were 1.2 km/h and 1.3 km/h, respectively.

Regression of km/d and km/h vs. fish length was not significant for smallmouth bass (km/d, $P > 0.4$, $r = 0.143$, 33 d.f.; km/h, $P > 0.5$, $r = 0.044$, 18 d.f.) or rock bass (km/d, $P > 0.5$, $r = 0.028$, 9 d.f.; km/h, $P > 0.4$, $r = 0.255$, 9 d.f.). However, regression of km/d vs. days tracked was significant for smallmouth bass ($P = 0.007$) and nearly so for rock bass ($P = 0.098$). Moderate negative correlations between km/d and days tracked (smallmouth bass, $r = 0.44$, 33 d.f.; rock bass, $r = 0.50$, 10 d.f.) reflect a propensity by fish released in the lake to move more during the initial days after release.

Table 2. Seasonal movement rates (km/d), short-term movement rates (km/h), seasonal temperatures (C) and seasonal activity levels (% active observations) of fish compared by species (smallmouth bass vs. rock bass) and habitat (Lake Ontario vs. tributaries).

Category	Seasonal Rate n $\bar{x} \pm SD$	Short-term Rate n $\bar{x} \pm SD$	Temperature n $\bar{x} \pm SD$	Activity n $\bar{x} \pm SD$
Lake Ontario				
Smallmouth Bass	35 0.7 $\pm$ 0.7	20 0.3 $\pm$ 0.2	35 15.1 $\pm$ 4.0	25 62 $\pm$ 14
Rock Bass	12 1.4 $\pm$ 1.5	11 0.4 $\pm$ 0.2	12 13.7 $\pm$ 4.0	8 62 $\pm$ 19
P Value	0.034	0.309	0.299	>0.5
Tributaries				
Smallmouth Bass	20 0.2 $\pm$ 0.2	19 0.1 $\pm$ 0.1	22 18.5 $\pm$ 2.7	18 35 $\pm$ 18
Rock Bass	5 0.5 $\pm$ 0.3	4 0.2 $\pm$ 0.1	5 16.0 $\pm$ 3.2	5 62 $\pm$ 24
P Value	0.004	0.056	0.087	0.010

Temperatures occupied by radiotagged bass in Lake Ontario did not differ between species (Table 2) and followed seasonal warming trends, with coldest temperatures in May and warmest temperatures in August. Colder temperatures recorded for smallmouth bass in Lake Ontario vs. tributaries (Table 1) were largely those of fish transported to the lake from late April through early June before the lake had warmed and stratified. Fish in tributaries generally did not move into Lake Ontario before June. Those that did often encountered temperature differences greater than 10 C and usually returned to the warmer tributary quickly.

Following initial movements, many fish settled in one location and established a home range, usually in an area of rocky structure in 4-8 m of water. Home areas occupied in Lake Ontario usually were so small that fish movements were detected only by variations in the signal strength of transmitters as fish moved; fish appeared to restrict their movements to a small patch of suitable habitat on the

lake bottom. Fifteen fish occupied areas near tributary discharges for brief periods and three smallmouth bass established residence areas just east of such discharges. Prevailing nearshore currents move eastward along Lake Ontario's southern shore (Saylor et al. 1981). Four smallmouth bass and two rock bass moved back and forth between home tributaries and the lake.

Observations of activity for smallmouth bass in Lake Ontario varied seasonally, decreasing from 76% in May to 31% in July and rising again in August and September (Figure 2). Rock bass did not show seasonal fluxes in activity levels. Smallmouth bass activity also varied by diel period; they were significantly more active ($P < 0.001$) during morning and afternoon periods than during the midnight period, with intermediate activity during dawn and dusk or twilight periods. Regression of smallmouth bass hourly activity levels vs. time of day was significant ($P < 0.001$, $r = 0.91$, 21 d.f.), indicating a strongly diurnal activity pattern (Figure 3). Diel activity for rock bass in Lake Ontario followed a similar pattern ($P = 0.001$, $r = 0.74$, 14 d.f.; Figure 3), although they were not tracked during the midnight period.

Table 3. Last known location of radiotagged smallmouth bass (n=44) and rock bass (n=12) in Lake Ontario and tributaries (n, %).

Category	Smallmouth Bass	Rock Bass
Transported to Lake Ontario	27	12
Resident at Release Site	4 (15%)	1 (8%)
Dispersed along South Shore	14 (52%)	6 (50%)
Returned to Home Tributary	7 (26%)	5 (42%)
Entered Another Tributary	2 (7%)	0
Re-entered Lake Ontario	0	2 (17%)
Released in Home Tributary	17	0
Resident in Tributary	9 (53%)	-
Entered Lake Ontario	8 (47%)	-

Tributaries. Eighty-eight percent (23 of 26) of fish displaced downstream or returning to home tributaries after release in the lake returned to original capture sites. Smallmouth bass occupied tributaries for 5-113 days. Over time, fish gradually restricted their movements to specific areas, usually in deeper water over rocky substrates downstream from shallow capture/spawning areas. Analysis of the movements of 15 smallmouth bass that established home ranges in tributaries revealed that 52% of all observations were

within 100 m of the center of activity and 73% were within 200 m (Figure 4). Excursions from 1.3 to 2.2 km followed by return to the home range center were observed for five fish on seven occasions (Figure 4). Of the five rock bass that returned to Eighteen Mile Creek for 6-77 days, one established a residence area and four left the creek or disappeared within 6-15 days after returning.

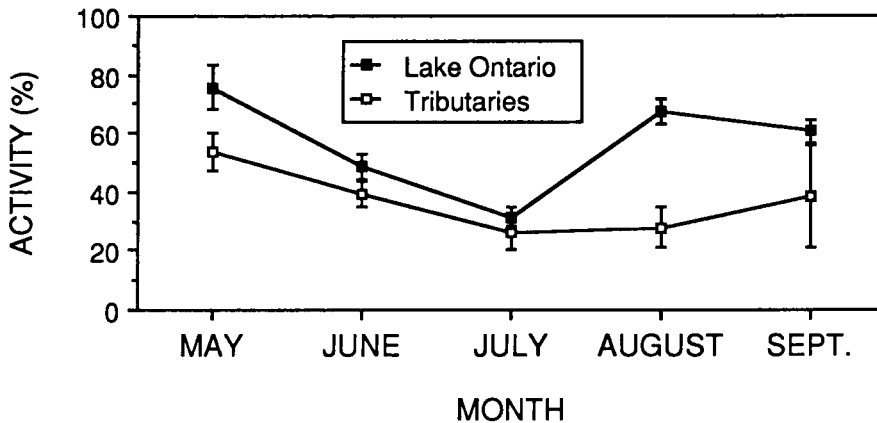


Figure 2. Activity levels of smallmouth bass in Lake Ontario and Sandy and Eighteen Mile Creeks by month. Based on 1690 observations of 44 fish.

As a result of rock bass dispersing from Eighteen Mile Creek sooner than smallmouth bass and generally not establishing residence areas within the creek, seasonal movement rates (km/d) in tributaries by rock bass were greater than for smallmouth bass (Table 2). Seasonal movements of rock bass did not differ between habitats, but seasonal rates for smallmouth bass were much lower in tributaries than in Lake Ontario (Table 1). Short-term movement rates (km/h) in tributaries did not differ between species (Table 2), but were significantly less than in Lake Ontario (Table 1).

Temperatures occupied by radiotagged fish in tributaries did not differ between species (Table 2), but were significantly higher than temperatures occupied in the lake (Table 1). This result reflects natural temperature differences between lake and tributary habitats. Due to their low volume and shallow depth, tributaries warmed earlier in the season, reached higher temperatures and exhibited greater diel temperature fluctuations than Lake Ontario.

Activity levels of rock bass in tributaries were higher than those observed for smallmouth bass (Table 2) and may have reflected their tendency to disperse rapidly after homing to tributaries. While activity levels of rock bass in tributaries did not differ from Lake

Ontario, smallmouth bass in tributaries were less active than in Lake Ontario (Table 1). Distinct diel activity patterns observed in Lake Ontario (Figure 3) were not observed in tributaries. Seasonal activity of smallmouth bass in tributaries was not significantly different by month; however, activity followed the same general pattern described for smallmouth bass in Lake Ontario (Figure 2). Rock bass did not occupy tributaries long enough to make seasonal comparisons by month.

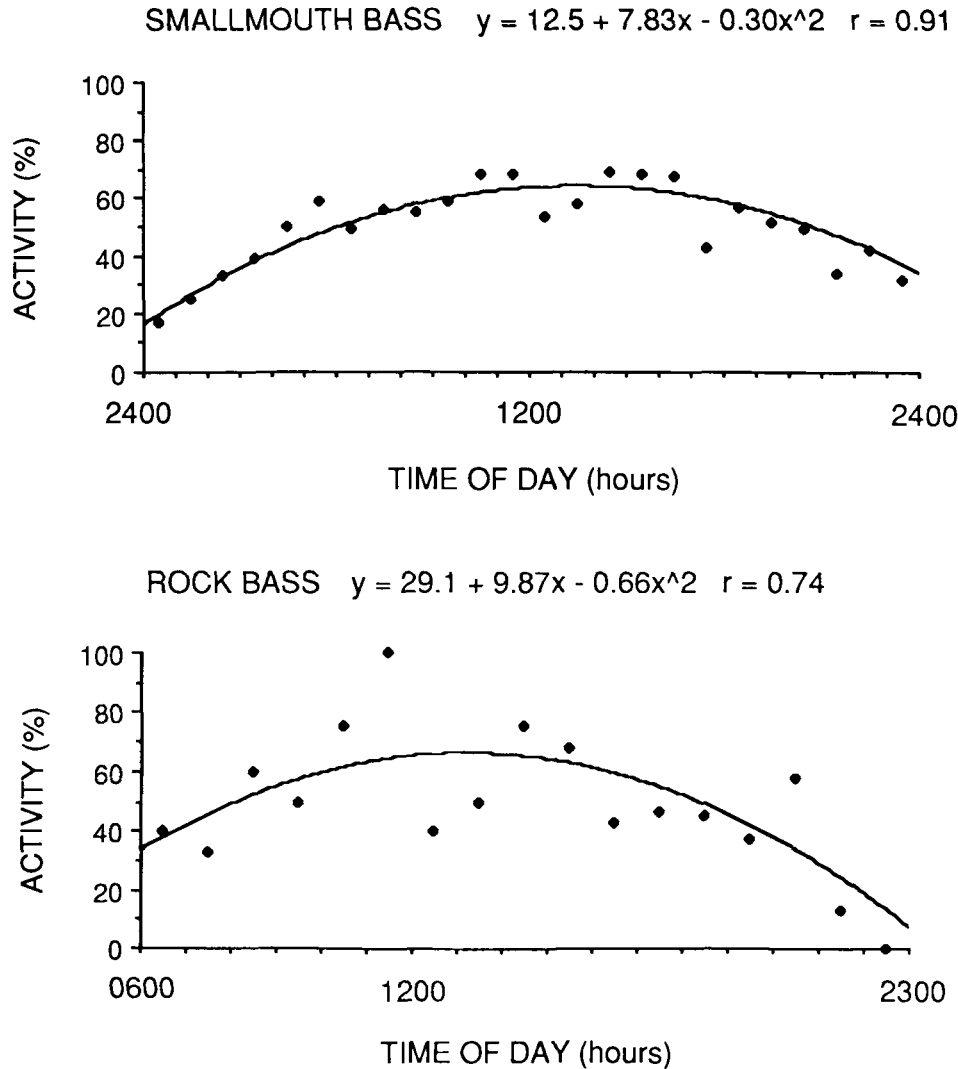


Figure 3. Activity levels of smallmouth bass (top) and rock bass (bottom) in Lake Ontario vs. time of day. Based on 1053 observations of 33 smallmouth bass and 127 observations of 12 rock bass.

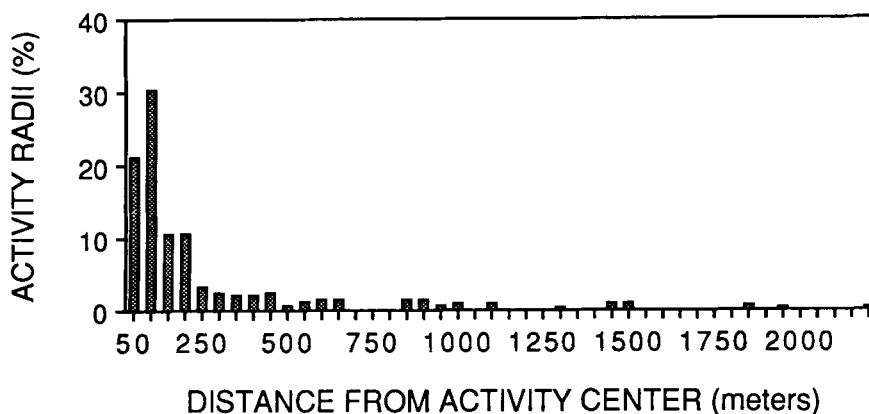


Figure 4. Cumulative frequency distribution of smallmouth bass locations relative to their centers of activity in tributaries. Based on 636 observations of 15 fish.

DISCUSSION

Activity. Consistent with previous studies, we found that smallmouth bass and rock bass were rarely active at night (Munther 1970, Emery 1973, Helfman 1981) and activity/movement was greater in spring than summer (Munther 1970, Peterson and Myhr 1976, Hubert and Lackey 1980). Smallmouth bass are considered crepuscular predators (Emery 1973) and are often reported to be most active during twilight periods (Hubbs and Bailey 1938, Keast and Welsh 1968, Emery 1973, Reynolds and Casterlin 1976, Helfman 1981). Although feeding activity may peak during twilight periods when prey are possibly more susceptible to predation (Emery 1973), we found that general activity/movement of adult fish was highest at mid-day, which was consistent with the studies of MacCrimmon and Robbins (1981) and MacLean et al. (1982). Fewer daytime movements observed in shallow, well-lit tributaries relative to Lake Ontario may have resulted from the tendency of smallmouth bass to seek shelter or over-head cover under high illumination conditions (Haines and Butler 1969) and from their preference for areas of low light-intensity (Reynolds and Casterlin 1976). MacLean et al. (1982) found that activity of smallmouth bass in Lake Erie increased with depth. This suggests that at lower illumination levels during the day in deeper lake waters, smallmouth bass may become less shelter-oriented and more active, as observed in Lake Ontario (Gerber 1987).

Temperature. Consistent with previous field studies (Hile and Juday 1941, Westman and Westman 1949, Hallam 1959), smallmouth

bass and rock bass occupied temperatures between 20-22 C when available. Although temperatures in parts of tributaries often exceeded 25 C in summer, bass rarely were observed at, and apparently avoided, these temperatures (Robbins and MacCrimmon 1977), despite demonstrating preferred temperatures from 26-31 C in the laboratory (Ferguson 1958, Peek 1965, Barans and Tubb 1973, Cherry et al. 1975, Coutant 1977, Reynolds and Casterlin 1978).

Smallmouth bass and rock bass appeared to move in relation to thermal structure, as do other fishes (cf. Brandt et al. 1980), and may be repelled by sudden temperature changes (Coutant 1975). Fish dispersing from tributaries in late spring and early summer before the lake had warmed and stratified generally returned to streams quickly (Gerber 1987). In summer some fish reappeared near shore during seiche events associated with large drops in nearshore temperatures (Gerber 1987). Wyman and Dischel (1984) found that impingement of smallmouth bass by power plant intakes on Lake Ontario's southern shore was significantly greater during or following upwelling events when the thermocline was in the vicinity of the intakes.

Movement. Most studies of smallmouth bass and rock bass movements and distribution have found these fish to inhabit waters less than 10 m deep (Webster 1954, Emery 1973, Helfman 1981, MacLean et al 1982, Storr et al. 1983), although habitation of deeper waters has been observed (Hile and Juday 1941, Webster 1954, MacLean et al. 1982). Our findings support both observations; in Lake Ontario smallmouth bass and rock bass routinely inhabited water < 6 m deep, but movement of fish to deeper water (sometimes apparently below the depth of signal detection) was fairly common in summer months (Gerber 1987).

In large lakes smallmouth bass are often characterized as existing in discrete populations (Stone et al. 1954, Webster 1954, Forney 1961). Mark-recapture and biotelemetry studies have revealed that smallmouth bass and rock bass often restrict their movements to small home areas (Scott 1949, Larimore 1952, Gerking 1953, Funk 1957, Fajen 1962, Munther 1970, Hubert and Lackey 1980). Several studies have demonstrated that displaced smallmouth bass and rock bass often return to the place of capture in breeding and non-breeding seasons and usually move more than non-displaced bass (Larimore 1952, Henderson and Foster 1957, Gerking 1959, Forney 1961, Hubert and Lackey 1980, Blake 1981, Pflug and Pauley 1983). Our results support these findings.

Other studies have documented the spawning migrations of smallmouth bass into streams from Lake Simcoe, Ontario, (Robbins and MacCrimmon 1977) and within the Columbia River system in southeastern Washington (Montgomery and Fickeisen 1978), as well as

the migrations of lake-spawning smallmouth bass (Kelso 1973) and rock bass (MacLean and Teleki 1977) in Long Point Bay, Lake Erie. Robbins and MacCrimmon (1977) found that stream-spawning smallmouth bass intermingled with lake-spawning, and possibly with other stream-spawning, smallmouth bass stocks on lake shoals during summer months, but suggested these groups were likely to be distinct genetically. Return of tagged fish to previously occupied sections of home tributaries in our studies, from release sites up to 10 km away, suggests that Lake Ontario may contain genetically distinct stocks of bass. Both smallmouth bass and rock bass preferentially chose home tributaries over other tributaries passed en route. However, preliminary results of allozyme frequency studies of Sandy and Eighteen Mile Creek smallmouth bass stocks, using starch gel electrophoresis, identified no polymorphic loci and thus provide no information regarding population differences (Brett et al. 1987).

Smallmouth bass that returned to home tributaries usually restricted their movements and many established distinct home ranges downstream from spawning areas. Residence in these areas ranged from a few weeks to several months, indicating that some fish use tributary habitats throughout summer, probably well after spawning, and may be permanent stream residents. However, many fish disappeared from, or were tracked leaving, tributaries during summer months; electrofishing in tributaries near spawning areas after June produced few adult fish. Robbins and MacCrimmon (1977) and Webster (1954) also found that most smallmouth bass had moved downriver by July.

Management Implications. In large lake systems, movements of smallmouth bass and rock bass to spawn in tributaries may be widespread. Although we were unable to identify population differences electrophoretically, bass stocks homing to tributaries may represent genetically distinct populations (Robbins and MacCrimmon 1977). The possible existence of homing populations in Lake Ontario and elsewhere (Robbins and MacCrimmon 1977) has implications for the management and enhancement of centrarchid stocks in large lakes. If populations spawning in tributaries are important contributors to lake-wide stocks, improving tributary habitats could have a dramatic effect on recruitment to existing populations. Eighteen Mile Creek embodies many of the problems that afflict Great Lakes tributaries and may restrict fish populations: it is a known point source of toxic contaminants (IJC 1981), suffers from extensive agricultural development and associated fluctuations in flows and turbidity, and has a dam that blocks upstream passage. However, before the implications of impaired tributaries on Great Lakes smallmouth and rock bass populations can be fully understood, the relative

contributions of stream and lake-spawning stocks to lake-wide recruitment must be determined.

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